STAT-650

Project Report

Title

Pricing, Ratings, and Popularity: A Statistical Analysis of Sephora Beauty Products

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Abstract/Summary

This project analyzes 8,770 beauty and personal care products listed on the Sephora website to explore what factors influence product pricing, customer ratings, and customer engagement. The dataset was sourced from Kaggle and contains product-level information including price, category, brand, customer ratings, number of reviews, love count, and marketing labels such as exclusive, online-only, and limited edition. Understanding what drives pricing and customer satisfaction in the beauty retail industry is valuable for both consumers and brands looking to make informed decisions in a highly competitive market. Three research questions were investigated using multiple linear regression, logistic regression, and hypothesis testing including independent samples t-test, one-way ANOVA, and Tukey HSD post-hoc analysis. The multiple linear regression model found that product category is the strongest predictor of price, with Perfume products priced significantly higher and Hair Styling Products priced significantly lower than other categories. Contrary to expectations, Sephora exclusive products were associated with lower prices after controlling for category. The logistic regression model for predicting high ratings achieved an AUC of 0.664, with product category again emerging as the dominant predictor while price and popularity metrics had very little practical effect. The hypothesis tests revealed that exclusive products receive significantly more customer favorites than regular products, but the most unexpected finding was that limited edition products had the lowest average love count of all three groups, directly contradicting the initial assumption that scarcity would drive higher engagement. These findings suggest that product category plays a central role in shaping both pricing and customer perception on Sephora, and that marketing labels like exclusivity and limited edition do not always lead to the outcomes that might be intuitively expected.

Introduction

Background and Motivation

The dataset used in this project was obtained from Kaggle and is titled "All Products Available on Sephora Website." It was scraped directly from the official Sephora website and contains information on beauty and personal care products listed for sale on the platform. The dataset has 9,168 rows and 21 columns, where each row represents one product listed on the Sephora website. The variables include both product-level details and customer engagement metrics such as brand, category, product name, size, price, value price, customer rating, number of reviews, love count, and binary indicators for

whether a product is sold online only, is a Sephora exclusive, is a limited edition release, or is a limited time offer. There are also columns capturing marketing flags, product details, usage instructions, and ingredient lists, which are more descriptive in nature and were not used for statistical modeling.

This dataset was chosen because it offers a real-world view into how a major beauty retailer structures and presents its products to consumers. The beauty and cosmetics industry is a massive and growing market, and platforms like Sephora serve as key places where brand positioning, pricing strategy, and customer feedback all intersect. Having both pricing data and customer engagement metrics in one place makes this dataset well-suited for applying regression and classification methods to answer questions about what drives product prices and ratings. It also allows for group comparison tests to explore whether marketing strategies like exclusivity or limited-edition releases are linked to higher consumer interest, as measured by the love count variable.

Literature Review

Research on the cosmetics industry consistently shows that brand image, price perception, and product quality are the main drivers of consumer purchase intention. Othman et al. (2022) conducted a regression-based study on cosmetic consumers in Malaysia and found that brand image, perceived price, and perceived quality together explained about 68.9% of the variation in purchase intention. Their findings showed that consumers are sensitive to price and tend to view higher-priced products with lower purchase intention unless they trust the brand. This directly connects to our first research question, where we use multiple linear regression to understand which product features, such as category and exclusivity, predict price on the Sephora platform.

The role of online ratings and reviews in shaping consumer behavior is also well-documented in e-commerce research. Medill Spiegel Research Center (Northwestern University) found that purchase likelihood consistently peaks for products with ratings between 4.0 and 4.7 stars, with reviews having an even stronger impact on higher-priced products, where displaying reviews increased conversion rates by up to 380% compared to lower-priced items. Additionally, research from Pattern Data Science showed that every 0.5-star improvement in a product's rating increases sales by approximately 3% across all channels. These findings strongly suggest that a threshold effect exists around the 4.0 rating mark, which directly motivates our second research question. We expect that features like number of reviews, love count, and price will help predict whether a Sephora product crosses that high-rating threshold, and we use logistic regression to test this.

The consumer psychology behind limited edition and exclusive products has also received academic attention. Kim and Johnson (2017), published in the *Journal of Retailing and Consumer Services*, examined the effect of limited-edition offers on luxury consumer perceptions and found that limited-edition labeling improves brand perception and drives stronger customer engagement, particularly among consumers motivated by exclusivity and self-presentation. This study found that scarcity marketing, including limited-edition releases, positively influenced consumer attitudes and behavioral intentions such as repurchase and word-of-mouth. This is relevant to our third research question, where we test whether Sephora products labeled as exclusive or limited edition actually receive statistically higher love counts compared to regular products, using a t-test and one-way ANOVA.

Research Goals and Hypotheses

The goal of this project is to use statistical modeling techniques to better understand what drives product pricing, customer ratings, and consumer engagement on the Sephora platform. Three research questions guide this analysis; each paired with a formal hypothesis.

Research Question 1: What product characteristics, such as category, online availability, and exclusivity status, predict the price of a Sephora beauty product?

The first goal is to identify which product-level features are significant predictors of price. I will use multiple linear regression with price as the numeric response variable and category, online_only, exclusive, and limited_edition as predictor variables.

- **H_0:** None of the product characteristics (category, online_only, exclusive, limited_edition) have a significant effect on product price.
- **H_1:** At least one product characteristic has a significant effect on product price.

Based on the literature, I expect that product category will be the strongest predictor of price, with categories like perfume and skincare commanding higher prices than general body care or accessories. I also expect exclusive products to be priced higher on average, since exclusivity is often associated with premium positioning.

Research Question 2: Can we predict whether a Sephora product is highly rated (rating of 4.0 or above) based on its price, number of reviews, love count, and product category?

The second goal is to classify products as highly rated or not using logistic regression. A binary response variable is created where 1 indicates a rating of 4.0 or above and 0 indicates otherwise. Model performance will be evaluated using a confusion matrix and prediction accuracy.

- **H_0:** Price, number of reviews, love count, and category do not significantly predict whether a product is highly rated.
- **H_1:** At least one of these variables significantly predicts whether a product is highly rated.

I expect that products with a higher number of reviews and higher love counts will be more likely to be highly rated, as greater engagement generally reflects stronger consumer satisfaction. Price may have a mixed effect, since both very cheap and very expensive products can attract extreme ratings in either direction.

Research Question 3: Do exclusive or limited-edition Sephora products receive significantly more customer "loves" than regular, non-exclusive products?

The third goal is to test whether marketing labels like exclusivity and limited-edition status are linked to higher consumer engagement, measured through the love count variable. We use an independent samples t-test and one-way ANOVA with a significance level of 0.05.

- **H_0:** There is no significant difference in mean love count between exclusive/limited edition products and regular products.
- **H_1:** Exclusive or limited-edition products have a significantly higher mean love count than regular products.

Based on prior research on scarcity marketing, we expect that products labeled as exclusive or limited edition will have higher love counts on average, as these labels tend to signal desirability and drive stronger consumer interest and engagement.

Data

Data Description

The dataset used in this project is the "All Products Available on Sephora Website" dataset, which was sourced from Kaggle (https://www.kaggle.com/datasets/raghadalharbi/all-products-available-on-sephora-website). It was scraped from the official Sephora website and accessed as a CSV file

downloaded directly from Kaggle. The dataset contains 9,168 rows and 21 columns, where each row represents a single beauty or personal care product listed for sale on the Sephora website. The variables used in this project are described in Table 1 below.

Variable	Type	Description
id	Numeric	Unique product identifier
brand	Categorical (324 levels)	Brand or manufacturer of the product
category	Categorical (143 levels)	Product type (e.g., Perfume, Moisturizers, Face Serums)
name	Text	Product name
size	Text	Product size or volume
rating	Numeric (1.0 – 5.0)	Average customer star rating
number_of_reviews	Numeric (count)	Total number of customer reviews
love	Numeric (count)	Number of times a product was favorited by customers
price	Numeric (USD)	Listed retail price on Sephora
value_price	Numeric (USD)	Original value price before any markdown
URL	Text	Product webpage link
MarketingFlags	Boolean	Whether the product has a marketing flag
MarketingFlags_content	Text	Describes the marketing flag (e.g., "online only")
options	Text	Available size or shade options
details	Text	Product description
how_to_use	Text	Usage instructions
ingredients	Text	Full ingredient list
online_only	Binary (0/1)	1 = sold online only, 0 = available in-store and online
exclusive	Binary (0/1)	1 = Sephora exclusive product, 0 = not exclusive
limited_edition	Binary (0/1)	1 = limited edition release, 0 = regular product
limited_time_offer	Binary (0/1)	1 = limited time offer, 0 = standard listing

Data Review and Cleaning

When the dataset was first loaded, a check for missing values showed that all 21 columns had zero missing values, and no duplicate rows were found. However, 398 products had a rating of 0.0, which indicates that those products had not yet received any customer ratings. Since a rating of 0 does not represent a real rating on the 1 to 5 scale, these rows were removed. After removing the unrated products, the dataset had 8,770 rows remaining. Since rating is the response variable in my second research question, I chose to remove these rows rather than impute a default value. Imputing a value like the mean or median for an unrated product would introduce artificial data into the variable I am trying to predict, which could bias the model results. Removing them was the more appropriate choice given that rating of 0 carries no real information about customer satisfaction. Additionally, nine columns that were not useful for statistical analysis were dropped: URL, MarketingFlags, MarketingFlags_content, options, details, how_to_use, ingredients, size, and name. These columns contained long text strings such as ingredient lists and usage instructions that do not contribute to the modeling objectives of this project. After these steps, the final cleaned dataset has 8,770 rows and 12 columns, which are: id, brand, category, rating, number_of_reviews, love, price, value_price, online_only, exclusive, limited_edition, and limited_time_offer. Since there were no missing values in the remaining analytical columns, no further imputation was needed.

Data Summary

Summary statistics for the five numeric variables in the cleaned dataset are as follows. The variable price ranges from $2.00 to $549.00 with a mean of $49.82 and a standard deviation of $46.88, indicating that most products are priced under $100 but the distribution is right-skewed with some high-end outliers. The value_price variable has a nearly identical distribution, with a mean of $51.33, suggesting that most products are not heavily discounted from their original value. The rating variable ranges from 1.0 to 5.0 with a mean of 4.17 and a very low standard deviation of 0.55, meaning most products on Sephora are rated quite highly, with the 25th percentile already at 4.0. The number_of_reviews variable is heavily right-skewed, with a mean of 294.88 but a median of only 53, and a maximum of 19,000, showing that a small number of products receive the vast majority of reviews. Similarly, the love count has a mean of 16,984 but a median of 5,200 and a maximum of 1,300,000, again indicating a highly skewed distribution. For the categorical and binary variables, the most represented brand in the cleaned dataset is SEPHORA COLLECTION with 465 products, and the most common category is Perfume with 656 products. Of the 8,770 products, 1,953 are sold online only, 2,321 are Sephora exclusives, 722 are limited edition, and only 3 are listed as a limited time offer.

The figures below show the association between each response variable and its explanatory variables. In the first plot, categories like Hair Straighteners and Blowdryers, Curling Irons, and Anti-Aging have the highest average prices, suggesting that product category is likely to be an important predictor of price. In the second plot, approximately 7,100 products out of 8,770 are classified as highly rated (rating of 4.0 or above), compared to about 1,670 that are not, showing a class imbalance that will be noted during the logistic regression analysis. In the third plot, exclusive products have the highest average love count at around 20,500, followed by regular products at about 16,000, while limited edition products have the lowest average love count at approximately 8,000, which is somewhat contrary to what was initially expected. Figure 2 further shows the association between the predictors and high rating status. Both price and love count show similar median distributions across highly rated and non-highly-rated products, though highly rated products show a greater number of extreme outliers in love count, suggesting that a small number of very popular products tend to be highly rated. Note that Figure 1 shows the top 10 categories by average price. The regression model in RQ1 uses the top 10 categories by product count instead, to ensure enough observations per group.

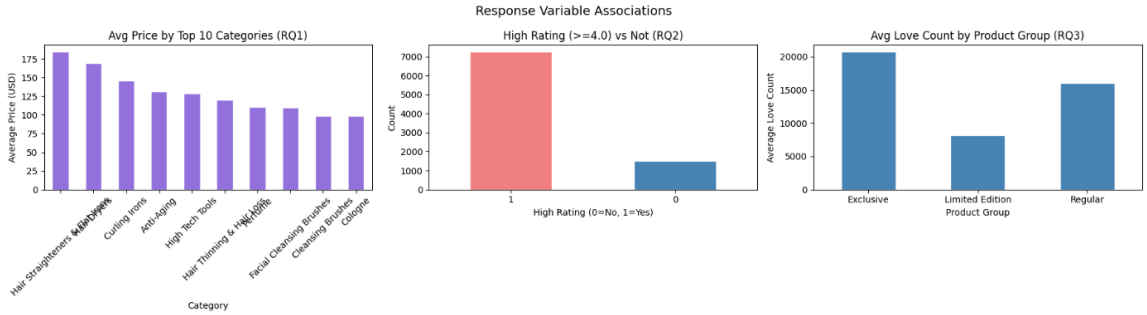


Figure 1: Response Variable Associations - Average Price by Top 10 Categories (RQ1), High Rating Distribution (RQ2), Average Love Count by Product Group (RQ3)

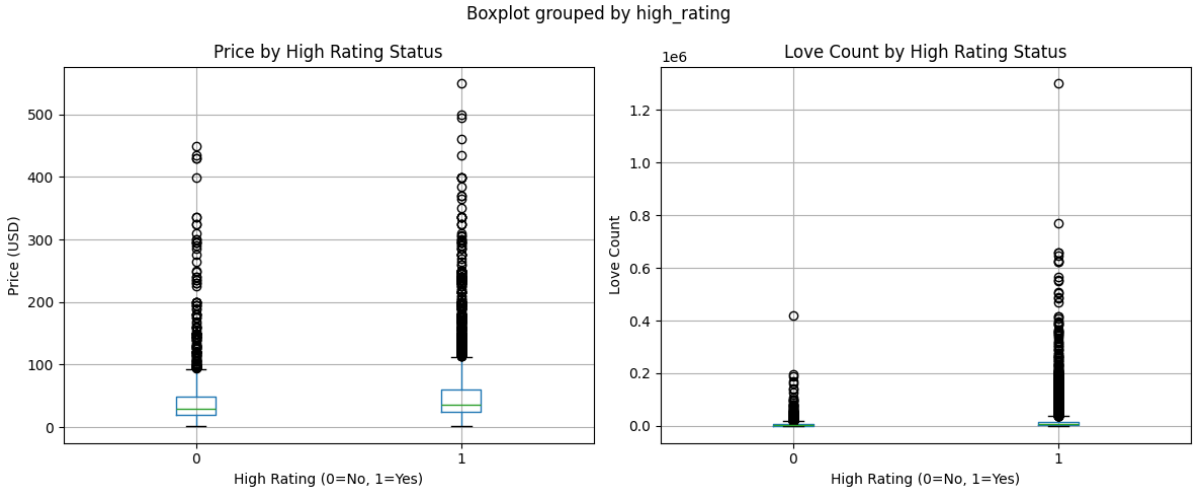


Figure 2: Association between predictors and high rating status - (Left) Price by High Rating Status, (Right) Love Count by High Rating Status. Both plots show the distribution of each predictor across highly rated (1) and not highly rated (0) products

Methodology

Research Question -1

What product characteristics, such as category, online availability, and exclusivity status, predict the price of a Sephora beauty product?

Model and Justification

To answer the first research question, I used Multiple Linear Regression (MLR) with price as the numeric response variable. MLR is appropriate here because the response variable is continuous and I am interested in understanding how multiple predictor variables together explain variation in product price. The predictors used are category_grouped, online_only, exclusive, and limited_edition. Since category originally had 143 unique levels, I grouped the top 10 most frequent categories by product count and assigned all remaining categories to an "Other" group, resulting in 11 levels total. This reduced the number of dummy variables from 142 to 10, making the model more interpretable and stable. The category_grouped variable was then dummy encoded using pd.get_dummies() with the first level dropped to avoid multicollinearity, leaving 10 category dummy variables. The final model has 13 predictors and was fit using Ordinary Least Squares (OLS) via the statsmodels library, which provides full coefficient estimates, standard errors, t-statistics, and p-values.

Hypotheses and Significance Level

- **H_0:** None of the product characteristics have a significant effect on product price.
- **H_1:** At least one product characteristic has a significant effect on product price.

The significance level used is $\alpha = 0.05$. Individual predictors are considered statistically significant if their p-value is less than 0.05. The overall model significance is evaluated using the F-statistic and its associated p-value.

Model Validation and Performance

The overall model is statistically significant with an F-statistic of 209.7 and a p-value of 0.00, meaning I reject the null hypothesis that none of the predictors affect price. The model explains 23.7% of the variation in price ($R^2 = 0.237$, Adjusted $R^2 = 0.236$). On the held-out test set (20% of data), the model achieved an R^2 of 0.2396 and an RMSE of $41.43, meaning predictions are off by about $41 on average. Given that prices range from $2 to $549 with a mean of $49.82, this level of error is relatively large, suggesting the model captures the general trend but misses a substantial portion of the variation in price.

To validate the model assumptions, I examined the residuals vs. fitted values plot and the Q-Q plot of residuals. The residuals vs. fitted plot shows a clear fan-shaped pattern, indicating heteroscedasticity - the variance of residuals increases as fitted values increase. This means the equal variance assumption of OLS is violated. The Q-Q plot shows that residuals deviate heavily from the normal line at both tails, indicating the residuals are not normally distributed and are right-skewed (Skew = 3.498, Kurtosis = 24.037 as reported in the OLS summary). These violations suggest the model results should be interpreted with caution, and that a log transformation of the response variable price may improve model fit.

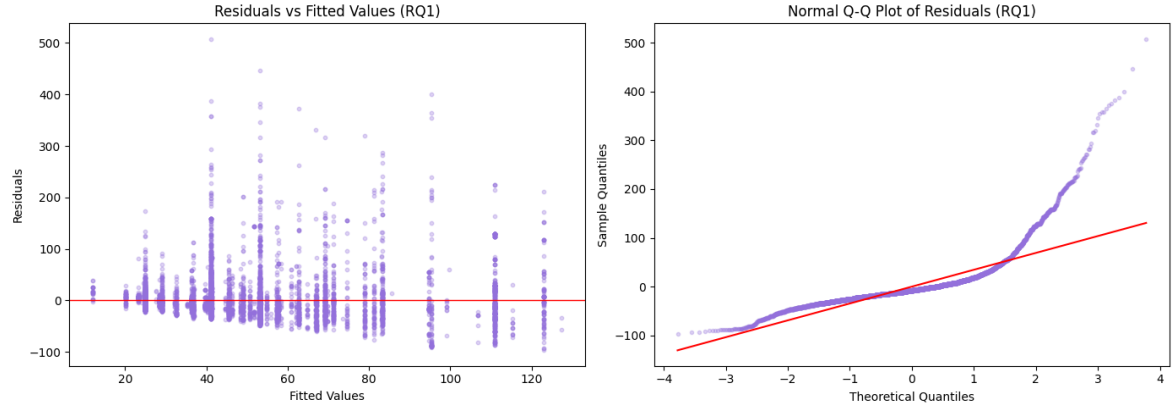


Figure 3: Original Model Residuals

Model Revision

After examining the residual plots from the initial model, I found evidence of heteroscedasticity and non-normality in the residuals, with a skewness of 3.498 and kurtosis of 24.037. To address this, I revised the model by applying a log transformation to the response variable, replacing price with log(price). This is a standard approach when the response variable is right-skewed, as it stabilizes the variance of residuals and brings the distribution closer to normality.

The revised model shows meaningful improvement across all diagnostics. The R^2 increased from 0.237 to 0.269 and the F-statistic increased from 209.7 to 248.0, indicating better overall model fit. More importantly, the residual diagnostics improved substantially. The residuals vs. fitted plot no longer

shows a clear fan shape, and the Q-Q plot follows the diagonal line much more closely through the middle range of the data, with only moderate deviation at the tails. The skewness dropped dramatically from 3.498 to 0.082, and kurtosis reduced from 24.037 to 4.338, confirming that the log transformation significantly improved the normality of residuals. All predictors that were significant in the original model remain significant in the revised model. The revised log(price) model is therefore used as the final model for RQ1, and all results and interpretations in the Results section are based on this model.

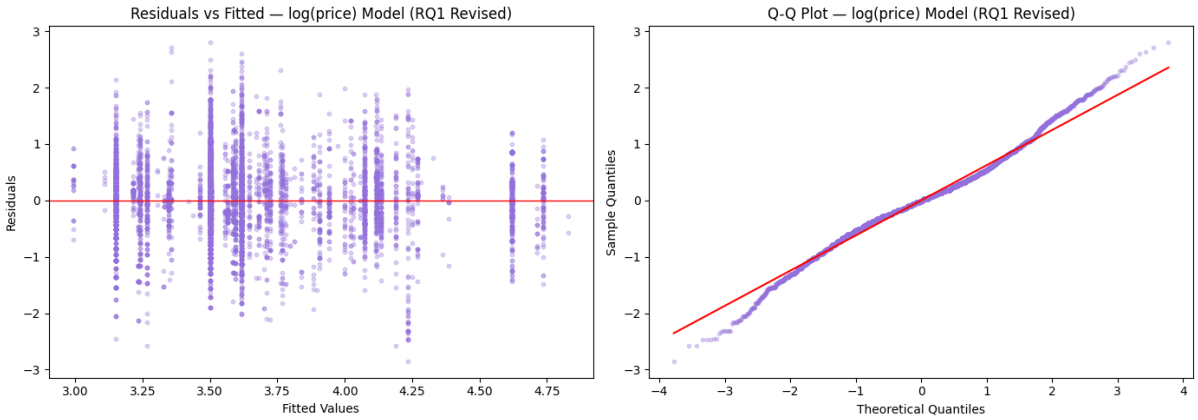


Figure 4: Revised log(price) Residuals

Research Question 2 –

Can we predict whether a Sephora product is highly rated (rating of 4.0 or above) based on its price, number of reviews, love count, and product category?

Model and Justification

To answer the second research question, I used Logistic Regression with a binary response variable high_rating, defined as 1 if a product has a rating of 4.0 or above and 0 otherwise. Logistic regression is appropriate here because the response variable is binary and I am interested in modeling the probability that a product is highly rated given its features. The predictors used are price, number_of_reviews, love, and category_grouped. The same category grouping from RQ1 was applied, keeping the top 10 categories and grouping the rest as "Other," resulting in 10 dummy variables after dropping the first level. The final model has 13 predictors and was fit using Maximum Likelihood Estimation (MLE) via the statsmodels Logit function, which provides coefficient estimates, standard errors, z-statistics, and p-values. The model was also fit using sklearn's LogisticRegression on an 80/20 train-test split with stratify=y to preserve the class proportions across both sets.

Before fitting the model, I checked the class distribution of the response variable. Of the 8,770 products in the cleaned dataset, 7,252 (82.7%) are highly rated and 1,518 (17.3%) are not, indicating a significant class imbalance. This imbalance is noted as a limitation and its effect on model performance is discussed in the validation section below.

Hypotheses and Significance Level

- H_0: Price, number of reviews, love count, and product category do not significantly predict whether a product is highly rated.
- H_1: At least one of these variables significantly predicts whether a product is highly rated.

The significance level used is $\alpha = 0.05$. Individual predictors are considered statistically significant if their p-value is less than 0.05.

Model Validation and Performance

The overall model is statistically significant with a Log-Likelihood Ratio p-value of 1.83e-67, meaning I reject the null hypothesis that none of the predictors affect the probability of a high rating. The Pseudo R^2 (McFadden's) is 0.044, which is low but not unusual for logistic regression on real-world data - values between 0.02 and 0.40 are considered acceptable depending on the application.

On the held-out test set, the model achieved an overall accuracy of 82.67%. However, examining the confusion matrix reveals a critical issue driven by the class imbalance. The model predicted every single product as highly rated (class 1) - it correctly classified all 1,450 high-rated products but misclassified all 304 non-highly-rated products as high-rated. This means the model has a recall of 1.00 for class 1 but 0.00 for class 0, and a precision of 0.00 for class 0. The macro average F1-score is only 0.45, confirming that the high overall accuracy is misleading and is simply a result of the model defaulting to the majority class.

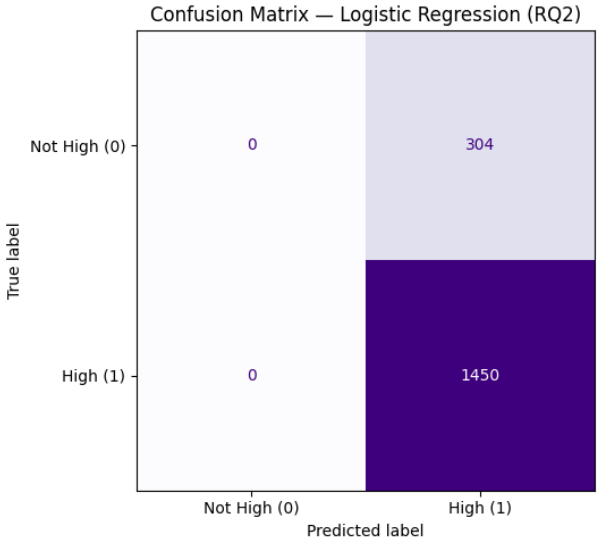


Figure 5: RQ2 Confusion Matrix

This performance is a direct consequence of the 82.7%/17.3% class imbalance. A classifier that simply predicts "high rating" for every product would also achieve 82.7% accuracy, which is essentially what the model is doing.

Model Revision

Given that the original model predicted every product as highly rated and failed entirely to identify the minority class, I revised the logistic regression model by setting class_weight='balanced' in sklearn's Logistic Regression. This parameter automatically adjusts the penalty for misclassifying each class inversely proportional to its frequency - the minority class (not highly rated, 17.3%) receives a much higher misclassification penalty than the majority class (highly rated, 82.7%), forcing the model to actually learn to identify both classes rather than defaulting to the majority.

The revised model shows a meaningful trade-off. Overall accuracy dropped from 82.67% to 51.31%, but the model now correctly identifies 234 out of 304 non-highly-rated products (recall of 0.77 for class 0), compared to 0 in the original model. The precision for class 0 improved from 0.00 to 0.23. For the

highly-rated class, recall dropped from 1.00 to 0.46, meaning the model now misses more high-rated products in exchange for catching more non-high-rated ones. The macro average F1-score improved from 0.45 to 0.48, and the weighted average F1-score is 0.57. While neither model performs strongly, the balanced model is more informative because it at least distinguishes between the two classes. The class imbalance in the dataset (82.7% vs 17.3%) is the primary limitation of this analysis and is discussed further in the Conclusion.

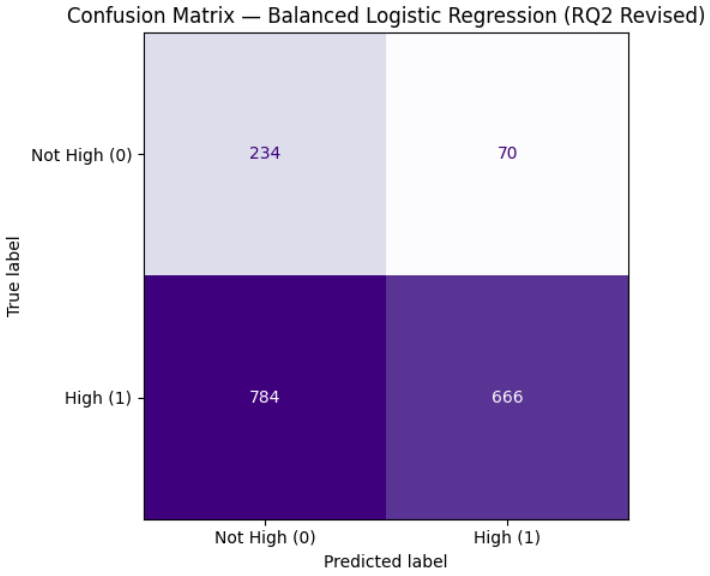


Figure 6: RQ2 Revised Confusion Matrix - Balanced Model

Research Question 3 –

Do exclusive or limited-edition Sephora products receive significantly more customer "loves" than regular, non-exclusive products?

Model and Justification

To answer the third research question, I used two statistical tests: an independent samples t-test and a one-way ANOVA. The response variable is love, which represents the number of times a product was favorited by customers on Sephora. Products were divided into three mutually exclusive groups based on their marketing label: Exclusive (exclusive = 1), Limited Edition (exclusive = 0 and limited_edition = 1), and Regular (both = 0). The group sizes are 2,321 Exclusive products, 338 Limited Edition products, and 6,111 Regular products.

The t-test was used to directly compare Exclusive and Regular products, as these are the two largest and most practically meaningful groups. The one-way ANOVA was then used to test all three groups simultaneously, since running multiple separate t-tests would inflate the false positive rate. These tests are appropriate here because the response variable is numeric and I am comparing means across independent groups.

Assumption Check and Model Revision

Before running the tests, I checked whether love count was normally distributed within each group using the Shapiro-Wilk test, since both t-test and ANOVA assume approximate normality. The results showed that raw love count is severely non-normal in all three groups (all p-values = 0.000), which is

consistent with the right-skewed distribution observed in the Data section, where the mean love count across all products was 16,984 but the median was only 5,200.

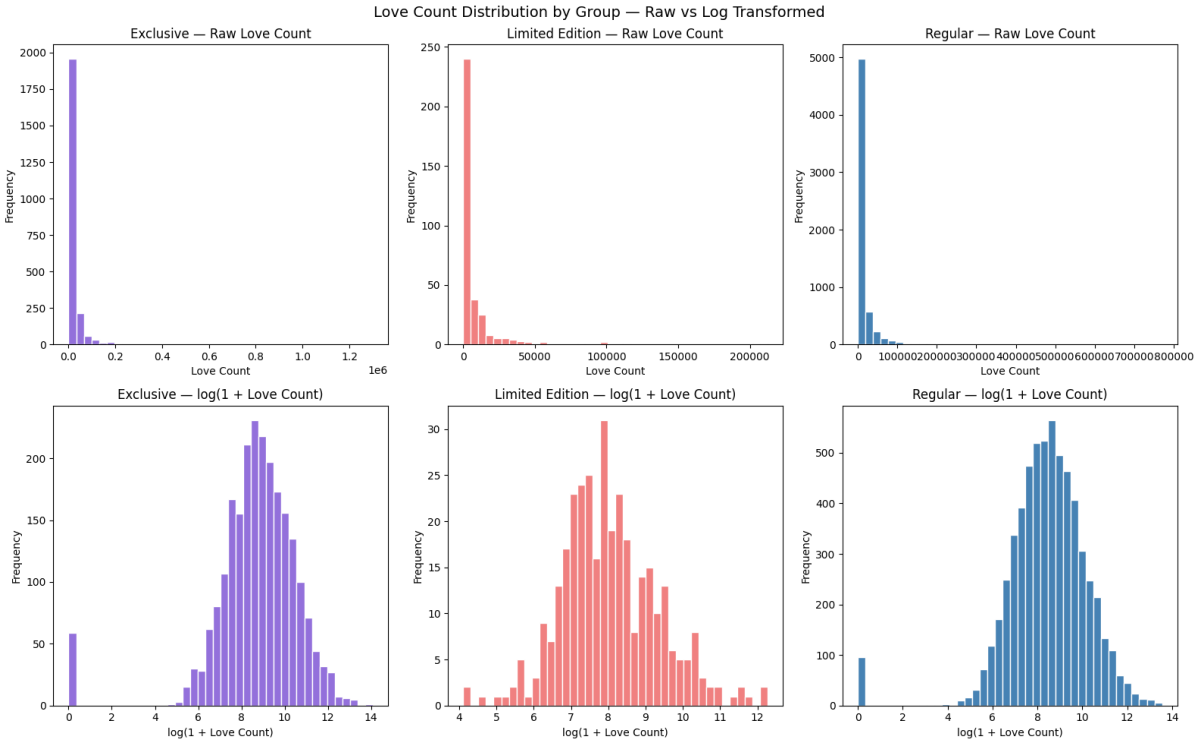


Figure 7: Love Count Distribution by Group - Raw vs Log Transformed

To address the violation of normality, I applied a log transformation using log(1 + love) to the love count variable. The addition of 1 before taking the log handles any products with a love count of 0, since log(0) is undefined. After transformation, the Shapiro-Wilk test still returned significant results for Exclusive ($p = 0.000$) and Regular ($p = 0.000$), and a p-value of 0.004 for Limited Edition. However, the W-statistics improved substantially from values as low as 0.35 to between 0.84 and 0.99, and the histograms of the log-transformed data show visually bell-shaped distributions for all three groups. Given the large sample sizes (n = 2,321, 338, and 6,111), the Central Limit Theorem ensures that the sampling distributions of the means are approximately normal even if individual observations are not perfectly normal. All subsequent tests were therefore conducted on the log-transformed love count variable.

Hypotheses and Significance Level

For the t-test:

- H0: The mean log love count for Exclusive products is equal to the mean log love count for Regular products
- H1: The mean log love count for Exclusive products is different from the mean log love count for Regular products

For the one-way ANOVA:

- H0: The mean log love count is equal across all three groups (Exclusive, Limited Edition, and Regular)
- H1: At least one group has a significantly different mean log love count

The significance level used for all tests is alpha = 0.05.

Model Validation and Performance

The Welch's independent samples t-test comparing Exclusive and Regular products produced a t-statistic of 5.28 and a p-value of 0.000, which is less than 0.05. I therefore reject the null hypothesis and conclude that there is a statistically significant difference in mean log love count between Exclusive and Regular products. Exclusive products have a higher mean log love count (8.68) compared to Regular products (8.43).

The one-way ANOVA across all three groups produced an F-statistic of 26.38 and a p-value of 0.000, confirming that at least one group mean differs significantly from the others. To determine exactly which pairs of groups differ, I ran the Tukey HSD post-hoc test. The results show that all three pairwise comparisons are statistically significant at alpha = 0.05. Exclusive products have significantly higher mean log love count than Limited Edition products (mean difference = 0.657, p = 0.000) and significantly higher than Regular products (mean difference = 0.250, p = 0.000). Limited Edition products have significantly lower mean log love count than Regular products (mean difference = -0.407, p = 0.000). This finding is notable because it shows that Limited Edition products have the lowest customer engagement of all three groups, which is contrary to what was initially hypothesized.

As an additional validation step, I ran the Kruskal-Wallis test, which is the non-parametric equivalent of one-way ANOVA and does not require the normality assumption at all. The Kruskal-Wallis test produced an H-statistic of 128.57 and a p-value of 0.000, fully consistent with the ANOVA result. The agreement between both tests confirms that the findings are robust and not dependent on the normality assumption.

Results

Question-1- Multiple Linear Regression on Price

What product characteristics, such as category, online availability, and exclusivity status, predict the price of a Sephora beauty product?

How the response variable is affected by the predictors

The revised model using log(price) as the response variable showed that product category is the strongest driver of price on Sephora. Among all category predictors, Perfume had the largest positive coefficient (0.49, p = 0.000), meaning Perfume products are associated with significantly higher prices compared to the baseline category (Anti-Aging). Hair Styling Products had the most negative coefficient (-0.79, p = 0.000), meaning they are priced significantly lower than the baseline. Face Wash and Cleansers (-0.63, p = 0.000), Other (-0.63, p = 0.000), Rollerballs and Travel Size (-0.57, p = 0.000), and Face Masks (-0.55, p = 0.000) also showed significantly lower prices compared to Anti-Aging. Face Serums (p = 0.784) and Moisturizers (p = 0.277) were the only categories that did not differ significantly in price from the baseline.

Among the binary variables, being a Sephora exclusive product had a notable negative coefficient (-0.35, p = 0.000), meaning exclusive products are actually priced lower than non-exclusive ones after controlling for category. Online-only products had a small positive coefficient (0.12, p = 0.000), and limited edition products had a small positive coefficient (0.09, p = 0.001), both statistically significant.

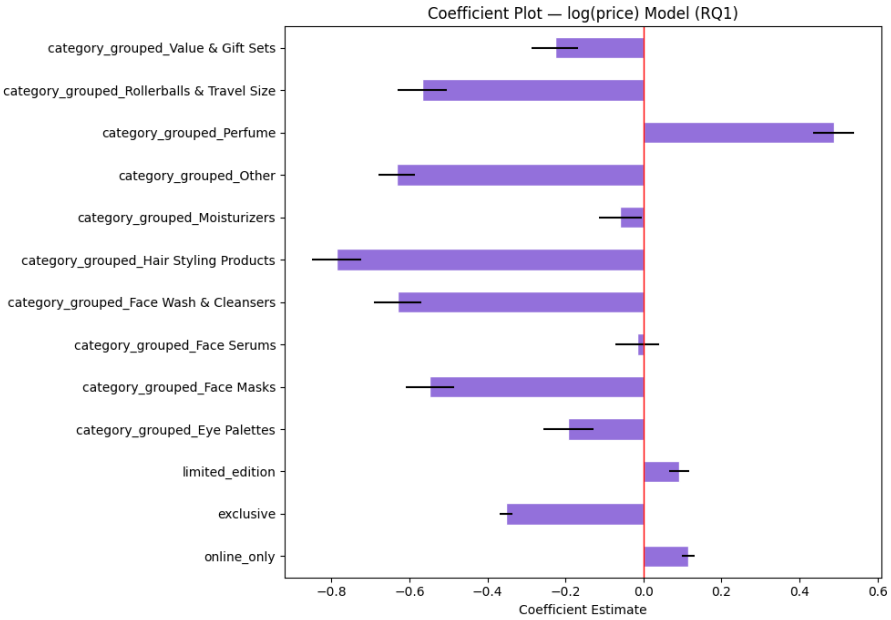


Figure 8: Coefficient Estimates - log(price) Revised Model (RQ1). Bars extending right of the red line indicate a positive effect on price. Bars extending left indicate a negative effect. Error bars show 95% confidence intervals.

Accuracy of the model

The overall model is statistically significant with an F-statistic of 248.0 and a p-value of 0.000. On the held-out test set (20% of data), the model achieved an R-squared of 0.271, meaning it explains about 27.1% of the variation in product price. The RMSE on the test set is $43.12 in the original price scale, meaning on average the model's price predictions are off by about $43. Given that most products are priced under $100 with a mean of $49.82, this is a moderate level of error.

Graph explanation

The actual vs predicted plot in Figure 9 shows the model performs well for products priced under $100, with points clustering near the red perfect-prediction line. For products priced above $100, the model consistently under-predicts, with predictions leveling off around $100 while actual prices go up to $549. This shows the model captures typical product pricing but struggles with high-end premium products.

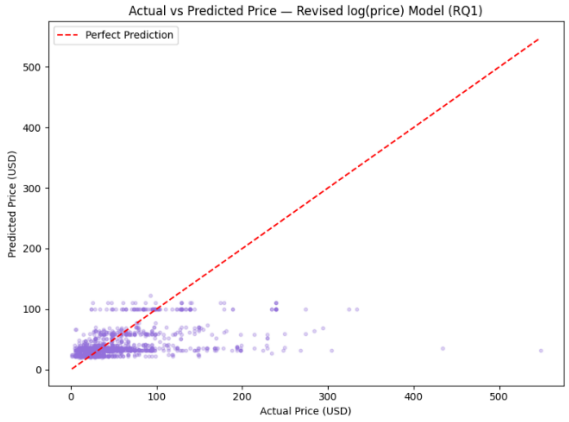


Figure 9: Actual vs Predicted Price in USD - Revised log(price) Model (RQ1). Points near the red dashed line indicate accurate predictions. Points below the line mean the model under-predicted the actual price.

Answer to Q1 and comparison to initial hypothesis

Yes, product characteristics do significantly predict price. Product category is the most important predictor, which matches my initial hypothesis. However, the finding that exclusive products are associated with lower prices, not higher prices, was unexpected and contradicts my initial assumption that exclusivity would command a price premium. The model's R-squared of 0.271 also suggests that a large portion of price variation is driven by factors not available in this dataset, such as brand prestige or product formulation.

Question - 2 - Logistic Regression on High Rating

Can we predict whether a Sephora product is highly rated (rating of 4.0 or above) based on its price, number of reviews, love count, and product category?

How the response variable is affected by the predictors

The logistic regression model showed that product category is the dominant predictor of whether a product is highly rated. All ten category dummy variables had statistically significant positive coefficients (p less than 0.05), meaning all included categories are more likely to be highly rated than the Anti-Aging baseline. The odds ratio plot in Figure 10 shows the magnitude of these effects. Face Wash and Cleansers had the highest odds ratio at approximately 5.94, meaning a product in that category is about 6 times more likely to be highly rated than an Anti-Aging product. Rollerballs and Travel Size had an odds ratio of approximately 5.41, and Perfume and Face Masks both had odds ratios of approximately 4.2.

In contrast, the continuous predictors had very small effects. Price had an odds ratio of approximately 1.003 (coef = 0.0029, p = 0.000), love count had an odds ratio of approximately 1.00002 (coef = 2.133e-05, p = 0.000), and number of reviews did not reach statistical significance at the 0.05 level (p = 0.056). While price and love count are technically significant, their odds ratios are so close to 1 that their practical effect is negligible.

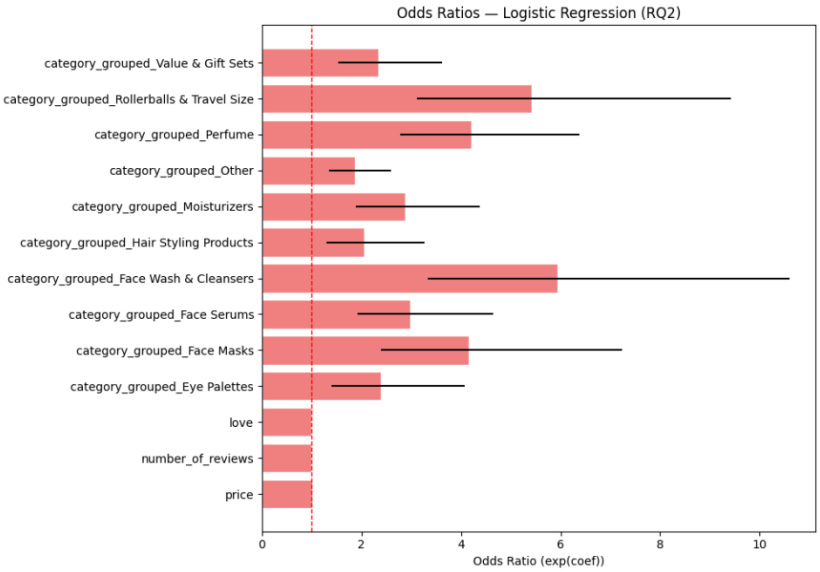


Figure 10: Odds Ratios - Logistic Regression (RQ2). Bars extending right of the red dashed line at 1.0 indicate that predictor increases the probability of being highly rated. Error bars show 95% confidence intervals.

Accuracy of the model

The original model achieved an accuracy of 82.67% but predicted every product as highly rated, making it practically useless. After applying class weighting to address the 82.7%/17.3% class imbalance, the balanced model achieved an accuracy of 51.31%, a macro average F1-score of 0.48, and a weighted F1-score of 0.57. The AUC score of 0.664 indicates the model has moderate ability to distinguish between highly rated and not highly rated products, performing better than random guessing (AUC = 0.5) but below what would be considered a strong classifier (AUC above 0.8).

Graph explanation

The ROC curve in Figure 11 plots the true positive rate against the false positive rate at different classification thresholds. The purple curve sits above the red dashed diagonal line (which represents random guessing) across most of the range, confirming the model provides some useful discriminatory ability. The AUC of 0.664 is the area under this curve and summarizes the model's overall ability in a single number.

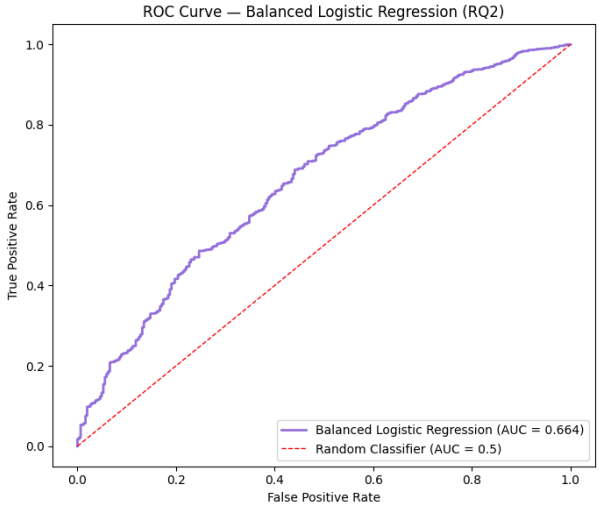


Figure 11: ROC Curve - Balanced Logistic Regression (RQ2). The purple curve shows model performance at all classification thresholds. The red dashed line shows what a random classifier would look like. The closer the curve is to the top left corner, the better the model.

Answer to Q2 and comparison to initial hypothesis

Product category is the most useful predictor of whether a product is highly rated. Price, number of reviews, and love count have very little practical influence on high rating status. My initial assumption that more popular and more reviewed products would be more likely to be highly rated is not supported by the data. The limited AUC of 0.664 also suggests that predicting rating from product features alone is inherently difficult, and the class imbalance in the dataset (82.7% highly rated) was the primary challenge for this model.

Question - 3 - t-test and ANOVA on Love Count

Do exclusive or limited-edition Sephora products receive significantly more customer "loves" than regular, non-exclusive products?

How the response variable is affected by the predictors

The Welch's independent samples t-test comparing Exclusive and Regular products on log-transformed love count produced a t-statistic of 5.28 and a p-value of 0.000, which is less than the significance level

of 0.05. Exclusive products had a higher mean log love count of 8.68 compared to Regular products at 8.43, confirming that exclusive products receive significantly more customer favorites on average.

The one-way ANOVA across all three groups produced an F-statistic of 26.38 and a p-value of 0.000, confirming that at least one group mean is significantly different from the others. The Tukey HSD post-hoc test showed that all three pairwise comparisons are statistically significant. Exclusive products had significantly higher mean log love count than Limited Edition products (mean difference = 0.657, p = 0.000) and significantly higher than Regular products (mean difference = 0.250, p = 0.000). Limited Edition products had significantly lower mean log love count than Regular products (mean difference = -0.407, p = 0.000). The Kruskal-Wallis non-parametric test confirmed these results with an H-statistic of 128.57 and p = 0.000, validating the findings hold without relying on the normality assumption.

Accuracy of the model

For hypothesis tests like t-test and ANOVA, accuracy is not measured through R-squared or RMSE. Instead, the strength of the result is evaluated through the test statistics and effect sizes. All tests produced p-values of 0.000, which is well below alpha = 0.05, providing very strong statistical evidence for group differences. The consistency between ANOVA (parametric) and Kruskal-Wallis (non-parametric) results further confirms the reliability of the findings.

Graph explanation

The boxplot in Figure 12 shows that Exclusive products have the highest median log love count and the widest spread, while Limited Edition products have the lowest median and the most compact distribution. The mean bar chart with standard error bars on the right confirms the Exclusive group has the highest average log love count (approximately 8.68), followed by Regular (8.43), with Limited Edition clearly the lowest (8.03). The very small error bars across all three groups indicate highly reliable mean estimates given the large sample sizes.

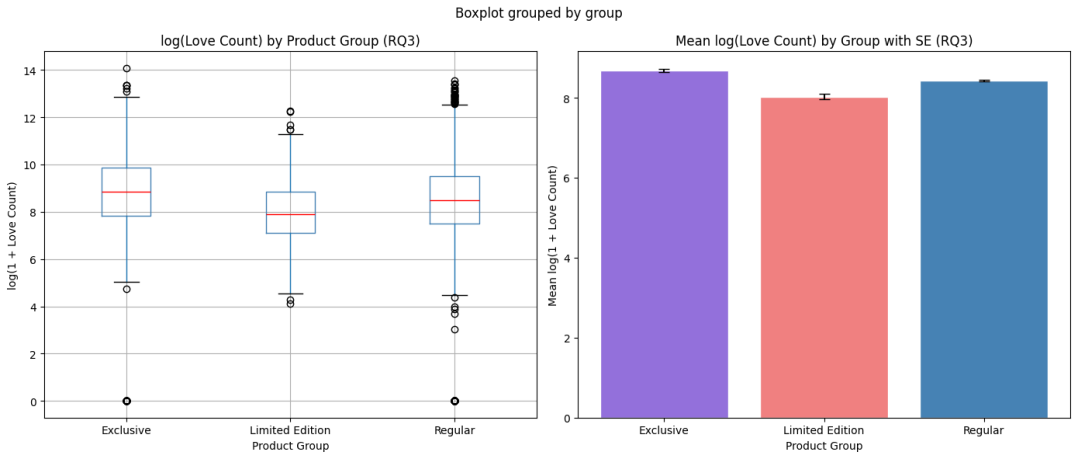


Figure 12: log(Love Count) by Product Group (RQ3). Left: Boxplot showing median, spread, and outliers for each group. Right: Mean log(Love Count) with standard error bars. Exclusive products have the highest average love count and Limited Edition products the lowest.

Answer to Q3 and comparison to initial hypothesis

Exclusive products do receive significantly more customer loves than Regular products, which partially supports my initial hypothesis. However, the most important finding from this analysis is that Limited Edition products have the lowest average love count of all three groups, significantly lower than both Exclusive and Regular products. This directly contradicts my initial expectation that scarcity and limited

availability would drive higher customer engagement for limited edition releases. This finding is discussed further in the Conclusion section.

Conclusion

Summary of Findings

This project analyzed 8,770 beauty and personal care products from the Sephora website to answer three research questions using multiple linear regression, logistic regression, and hypothesis testing.

For the first research question, I found that product category and marketing characteristics do significantly predict product price on Sephora. The multiple linear regression model on log-transformed price achieved an R-squared of 0.271 and an RMSE of $43.12 on the test set. Product category was the strongest predictor, with Perfume products priced significantly higher and Hair Styling Products priced significantly lower than the baseline. One unexpected finding was that Sephora exclusive products are actually associated with lower prices after controlling for category, which contradicted my initial assumption that exclusivity would command a price premium. Online-only and limited edition products were associated with slightly higher prices, but their effects were small.

For the second research question, I found that product category is the most useful predictor of whether a product is highly rated, while price, number of reviews, and love count have very little practical effect. The balanced logistic regression model achieved an AUC of 0.664, indicating moderate discriminatory ability. The severe class imbalance in the dataset, with 82.7% of products being highly rated, was the biggest challenge for this model and limited its ability to correctly identify non-highly-rated products even after applying class weighting.

For the third research question, I found statistically significant differences in love count across all three product groups. Exclusive products received significantly more customer favorites than both Regular and Limited Edition products. The most important and unexpected finding was that Limited Edition products had the lowest average love count of all three groups, significantly lower than both Exclusive and Regular products. This directly contradicted my initial hypothesis that scarcity and limited availability would drive higher customer engagement. This result was consistent across the Welch's t-test, one-way ANOVA, Tukey HSD post-hoc test, and the Kruskal-Wallis non-parametric test, all of which confirmed statistically significant group differences at alpha = 0.05.

Broader Impacts

The findings from this project have practical implications for beauty brands and retailers. The strong relationship between product category and both price and rating suggests that customers evaluate products primarily within their category context rather than across the entire store. For brands, this means that pricing strategy and quality expectations are highly category-specific and should not be treated uniformly across product types.

The finding that exclusive products are priced lower than expected challenges a common assumption in retail that exclusivity justifies higher prices. It is possible that Sephora uses exclusivity as a strategy to attract customers with affordable exclusive offerings rather than to charge premium prices. For retail analysts and brand managers, this suggests that the relationship between exclusivity and pricing is more complex than assumed and may vary significantly across product categories.

The low performance of the logistic regression model for predicting high ratings (AUC = 0.664) suggests that what makes a product highly rated is not easily captured by observable product

characteristics like price, category, or popularity metrics. This has implications for product development teams who might otherwise rely on these easily measurable attributes to predict customer satisfaction.

The unexpected finding about limited edition products having lower love counts than regular products is particularly relevant for marketing teams. It suggests that the novelty or scarcity of a limited edition label does not automatically translate into higher customer engagement on Sephora, and that these products may actually underperform regular listings in terms of being favorited by customers.

Future Research

There are several directions that could extend and improve this analysis. First, the models in this project used only structural product features such as category, price, and binary marketing flags. Adding text-based features extracted from product descriptions, ingredient lists, or customer reviews using natural language processing could significantly improve the prediction of both price and rating. Sentiment analysis of reviews in particular could provide a much richer signal for predicting high ratings.

Second, the limited time offer variable had only 3 products flagged as 1 after cleaning, making it essentially unusable in this analysis. A dataset collected over a longer time period with more limited time offer products would allow for a proper analysis of how time-limited promotions affect customer behavior.

Third, the class imbalance problem in RQ2 (82.7% highly rated vs 17.3% not highly rated) fundamentally limited the logistic regression model. Future work could explore more advanced techniques for handling class imbalance, such as SMOTE (Synthetic Minority Oversampling Technique) or ensemble methods like Random Forest with class weighting, which may produce better classification performance.

Fourth, this analysis treated all products independently. A multilevel or hierarchical model that accounts for the nested structure of products within brands and brands within categories could better capture the brand-level effects on price, rating, and customer engagement that the current models cannot fully account for.

Finally, a longitudinal dataset tracking how love counts and ratings change over time for the same products would allow for causal analysis rather than the purely cross-sectional analysis conducted here, which can only identify associations and not causal relationships.

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Appendix

Appendix A: Correlation Matrix of Numeric Variables

The correlation matrix below shows the pairwise correlations between all five numeric variables in the cleaned dataset. Price and value_price are highly correlated (r = 0.98), which is expected since value_price represents the original retail value before any markdown. Number of reviews and love count show a moderate positive correlation (r = 0.74), suggesting that products with more reviews also tend to have more customer favorites. Rating shows very weak correlations with all other variables (r between -0.09 and 0.08), indicating that a product's price or popularity does not strongly predict its average rating.

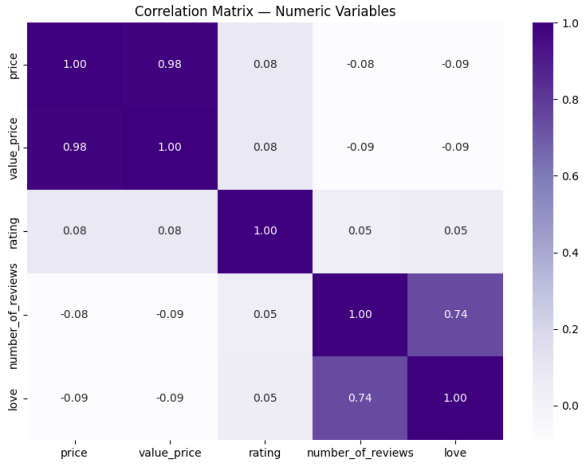


Figure A1: Correlation Matrix of Numeric Variables. Darker purple indicates stronger positive correlation. Values close to 0 indicate no linear relationship.

Appendix B: Full Category Distribution

category	count	category	count
Perfume	656	Eyebrow	103
Moisturizers	442	Concealer	100
Face Serums	378	Eye Brushes	91
Value & Gift Sets	316	Lotions & Oils	91
Face Wash & Cleansers	245	Toners	87
Face Masks	226	Face Oils	83
Rollerballs & Travel Size	220	Lip Balm & Treatment	81
Hair Styling Products	213	Mists & Essences	80
Eye Palettes	199	Scalp & Hair Treatments	79
Eye Creams & Treatments	189	Lip Gloss	79
Lipstick	188	Lip Balms & Treatments	78
Face Brushes	176	Eyeshadow	77
Shampoo	176	Hair Masks	76
Foundation	163	Face Sunscreen	74
Highlighter	161	Hair	69
Cologne	154	Blush	66
Mascara	153	Nail	56

Face Primer	143	Facial Peels	55
Mini Size	141	Sheet Masks	53
Conditioner	139	Hair Accessories	52
Candles & Home Scents	127	False Eyelashes	51
Eyeliner	123	Bronzer	50
Setting Spray & Powder	115	Skincare	48
Makeup	108	Bath & Shower	47
Body Lotions & Body Oils	108	Body Wash & Shower Gel	46
Beauty Supplements	105	Exfoliators	45
Perfume Gift Sets	104	Facial Cleansing Brushes	43
Dry Shampoo	43	Aftershave	13
Hair Oil	42	Eyelash Curlers	13
Face Sets	39	Self Tanners	12
For Body	39	Liquid Lipstick	12
Makeup Removers	37	Tweezers & Eyebrow Tools	12
Anti-Aging	37	Cellulite & Stretch Marks	11
Sponges & Applicators	36	Color Care	11
Blemish & Acne Treatments	34	BB & CC Creams	11
Curling Irons	32	Fragrance	10
Night Creams	30	Color Correct	10
Cheek Palettes	30	Decollete & Neck Creams	10
Body Mist & Hair Mist	29	Skincare Sets	10
Lip Liner	29	Mirrors & Sharpeners	9
Hair Spray	29	no category	9
Eye Primer	29	Eye Cream	9
Leave-In Conditioner	29	Bath & Body	9
Moisturizer & Treatments	28	Diffusers	7
Hair Brushes & Combs	28	Body Products	7
Hair Primers	27	Blotting Papers	7
Hair Straighteners & Flat Irons	27	Bath Soaks & Bubble Bath	6
Eye Sets	27	Hair Products	6
Brush Cleaners	26	Cologne Gift Sets	5
Hand Cream & Foot Cream	26	Lip Brushes	4
Facial Rollers	26	Hair Removal	4
Lip Sets	25	Shampoo & Conditioner	4
Makeup Palettes	25	Lip Plumper	3
Hair Dryers	24	Sunscreen	3
Shaving	24	After Sun Care	2
Holistic Wellness	23	Hair Removal & Shaving	2
BB & CC Cream	22	Lip Sunscreen	2
Face Wipes	22	Deodorant for Men	2
Candles	22	Hair Thinning & Hair Loss	1
Eye Masks	21	Hair Styling & Treatments	1
Scrub & Exfoliants	21	Cleansing Brushes	1
For Face	19	Curls & Coils	1
Brush Sets	19	High Tech Tools	1

Teeth Whitening	19	Wellness	1
Contour	18	Accessories	1
Body Sprays & Deodorant	18	Spa Tools	1
Lip Stain	18	Body Moisturizers	1
Face Wash	17	Lip Treatments	1
Body Sunscreen	17	Powder Brush	1
Deodorant & Antiperspirant	16	Lid Shadow Brush	1
Tinted Moisturizer	16		
Makeup Bags & Travel Cases	15		
Makeup & Travel Cases	15		

Appendix C: Full Brand Distribution

Brand	count	Brand	count
SEPHORA COLLECTION	465	Armani Beauty	55
CLINIQUE	219	HUDA BEAUTY	54
tarte	169	Caudalie	54
TOM FORD	154	Clarins	53
Fresh	137	Smashbox	52
Dior	137	CHANEL	52
Kiehl's Since 1851	124	Algenist	51
Lancôme	114	Gucci	51
Bumble and bumble	101	Dr. Jart+	50
MAKE UP FOR EVER	98	La Mer	48
Shiseido	92	VOLUSPA	48
Benefit Cosmetics	90	Jack Black	47
Kérastase	89	Briogeo	47
Moroccanoil	88	Dermalogica	47
FENTY BEAUTY by Rihanna	81	CLEAN RESERVE	47
philosophy	79	TOCCA	46
Too Faced	78	Dr. Dennis Gross Skincare	45
Bobbi Brown	74	Atelier Cologne	44
Urban Decay	74	REN Clean Skincare	44
Origins	73	DevaCurl	42
Jo Malone London	73	The Ordinary	42
Charlotte Tilbury	73	Kate Somerville	42
Givenchy	72	First Aid Beauty	42
Yves Saint Laurent	71	ALTERNA Haircare	42
KVD Vegan Beauty	70	shu uemura	41
Hourglass	69	Dr. Barbara Sturm	40
Anastasia Beverly Hills	68	The INKEY List	38
Peter Thomas Roth	67	Christophe Robin	38
MILK MAKEUP	67	Verb	37
Estée Lauder	65	Supergoop!	36
Laura Mercier	64	PAT McGRATH LABS	36
Living Proof	64	DOLCE&GABBANA	35
Perricone MD	63	L'Occitane	35

amika	63	Tatcha	35
Natasha Denona	60	Maison Margiela	35
Murad	59	BECCA	35
NARS	57	Josie Maran	34
bareMinerals	57	innisfree	34
IT Cosmetics	56	HERMÈS	33
Drybar	56	LANEIGE	33
Marc Jacobs Beauty	56	OUAI	33
NEST New York	55	Drunk Elephant	32
Guerlain	55	IGK	32
GLAMGLOW	32	Proactiv	22
OLEHENRIKSEN	31	Ralph Lauren	22
Viktor&Rolf	31	Ciaté London	22
T3	31	Virtue	22
AMOREPACIFIC	31	O&M	22
Tata Harper	30	Kopari	22
Herbivore	30	Skinfix	22
Kilian	30	Chloé	21
Acqua Di Parma	29	Dr Roebuck's	21
Dr. Brandt Skincare	28	AERIN	21
COOLA	28	Biossance	21
Primera	28	SK-II	20
Versace	28	PINROSE	20
belif	28	boscia	20
St. Tropez	28	By Rosie Jane	20
ELLIS BROOKLYN	27	Sephora Favorites	20
Mario Badescu	27	Rossano Ferretti Parma	19
NUDESTIX	27	Omorovicza	19
beautyblender	26	Sachajuan	19
rms beauty	26	KORA Organics	19
HUM Nutrition	26	Jouer Cosmetics	19
Melt Cosmetics	26	Prada	19
Maison Louis Marie	26	Floral Street	19
LANCER Skincare	26	Artis	19
Moon Juice	26	Sulwhasoo	19
FOREO	26	lilah b.	18
Rahua	25	The 7 Virtues	17
Juliette Has a Gun	25	KORRES	16
BURBERRY	25	dpHUE	16
INC.redible	25	Violet Voss	16
surratt beauty	25	SKYLAR	15
ILIA	24	Kaja	15
Eve Lom	24	PHLUR	15
Kitsch	24	Marvis	15
Farmacy	24	NuFACE	15
BVLGARI	24	Together Beauty	15

KEVYN AUCOIN	24	The Art of Shaving	15
BeautyBio	24	Saturday Skin	15
Deborah Lippmann	23	ghd	15
SUNDAY RILEY	23	Seed Phytonutrients	15
Sol de Janeiro	23	Anthony	14
Clarisonic Skincare	23	TWEEZERMAN	14
Viseart	23	Better Not Younger	14
Marc Jacobs Fragrances	23	FARSÁLI	14
NAILS INC.	23	Glow Recipe	14
Wander Beauty	22	Erborian	14
Youth To The People	14	RODIN olio lusso	8
BERDOUES	14	Reverie	8
Alpha-H	13	Comptoir Sud Pacifique	8
STELLAR	13	Playa	8
Erno Laszlo	13	John Varvatos	8
COVER FX	13	Narciso Rodriguez	8
Mizani	13	Obagi Clinical	8
Pretty Vulgar	13	Kosas	8
Grande Cosmetics	13	Nurse Jamie	8
Miu Miu	13	Paco Rabanne	8
Volition Beauty	13	Artist Couture	8
Bio Ionic	12	Mugler	7
Christian Louboutin	12	Olaplex	7
Mount Lai	12	Peace Out	7
Edible Beauty	12	Summer Fridays	7
TAN-LUXE	12	ROEN BEAUTY	7
Slip	12	Rosebud Perfume Co.	7
Bite Beauty	12	Koh Gen Do	7
Vitruvi	12	lululemon selfcare	7
Capri Blue	12	Lord Jones	7
Carolina Herrera	12	Elizabeth and James	7
PATRICK TA	12	Donna Karan	7
Velour Lashes	11	The Beauty Chef	7
Cinema Secrets	11	stila	7
Vital Proteins	11	Coal and Canary	6
Naturally Serious	11	Aether Beauty	6
FORM	11	rag & bone	6
Klorane	11	Shani Darden Skin Care	6
goop	10	Moschino	6
Juicy Couture	10	Buxom	6
Isle of Paradise	10	Montblanc	6
Indie Lee	10	Overose	6
Tory Burch	10	Black Up	5
KENZOKI	10	Prima	5
Qhemet Biologics	10	Tower 28 Beauty	5
LAWLESS	10	Aquis	5

StackedSkincare	9	GLO Science	5	s
Michael Kors	9	Fable & Mane	5	
Calvin Klein	9	DEREK LAM 10 CROSBY	5	
DOMINIQUE COSMETICS	9	alpyn beauty	5	
Lilly Lashes	9	WelleCo	5	
Good Dye Young	9	Skin Laundry	5	
Lab Series For Men	9	Rita Hazan	5	
PMD	9	Shaveworks	4	
dyson	8	KNC Beauty	4	
SOBEL SKIN Rx	8	Kenzo	4	
Pink Sugar	4	Etat Libre d'Orange	1	
Touch In Sol	4	High Beauty	1	
MAELYS Cosmetics	4	Flora + Bast	1	
trèStiQue	4	SHHHOWERCAP	1	
LASHFOOD	4	DL.MD	1	
Earth’s Nectar	4	Wishful	1	
Golde	4	Beachwaver	1	
Saint Jane Beauty	3	Taste Beauty	1	
dae	3	TokyoMilk	1	
Evian	3	Cellu-cup	1	
DERMAFLASH	3	bkr	1	
PLAY! by SEPHORA	3	MDSolarSciences	1	
Madam C.J. Walker Beauty Culture	3	Too Cool For School	1	
LAVANILA	3	Google	1	
Valentino	3	Conture	1	
Blinc	3	Lunar Beauty	1	
Antonym	3	Jillian Dempsey	1	
LXMI	3	Cocofloss	1	
Sakara Life	3	Eight & Bob	2	
ReFa	3	8Greens	2	
kate spade new york	3	LightStim	2	
Makeup Eraser	2	JIMMY CHOO	2	
Crushed Tonic	2	invisibobble	2	
BALENCIAGA	2	Commodity	2	
Winky Lux	2	YUNI	2	
DUO	2	Four Sigmatic	2	
Skin Inc Supplement Bar	2			
Bésame Cosmetics	2			